



CRISPR-knockout of β -kafirin in sorghum does not recapitulate the grain quality of natural mutants

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Abstract

Main conclusion When gene editing was applied to knockout beta-kafirin, there was a compensatory increase of gamma-kafirin which does not occur in domesticated null varieties, so enhanced grain quality was not achieved.

Abstract *Sorghum bicolor* is an important animal feedstock cereal crop throughout Australia and the southern United States, where its use as a food product is limited by issues with low calorific and nutritive value. Qualities such as reduced digestibility and low essential amino acid content are directly attributed to the kafirin grain storage proteins, the major components of protein bodies within the endosperm. Specifically, the β - and γ -kafirins have few protease cleavage sites and high levels of cysteine residues which lead to a highly cross-linked shell of intra- and inter-molecular disulphide linkages that encapsulate the more digestible α - and δ -kafirins in the core of the protein bodies. Naturally occurring β -kafirin mutants exist and are known to have improved grain quality, with enhanced protein contents and digestibility, traits which are often attributed to the lack of this cysteine-rich kafirin in the mature grain. However, when CRISPR/Cas9 editing was used to create β -kafirin knockout lines, there was no improvement to grain quality in the Tx430 background, although they did have unique protein composition and changes to protein body morphology in the vitreous endosperm. One explanation of the divergence in quality traits found the lines lacking β -kafirin are due to a drastic increase of γ -kafirin which was only found in the gene edited lines. This study highlights that in some germplasm, there is a level of redundancy between the peripheral kafirins, and that improvement of grain protein digestibility cannot be achieved by simply removing the β -kafirin protein in all genetic backgrounds.

Keywords Endosperm · Gene editing · Grain quality · Kafirin · Sorghum

Abbreviations

GE Gene edited line
PPIase Peptidyl-prolyl cis/trans isomerase
SEM Scanning electron microscopy

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Introduction

Cereal grains are the most important source of calories worldwide, with approximately 2.6 billion tonnes produced in 2018 (FAO 2018). In developed countries, close to 60% of calories are obtained from cereals, with this reaching up to 80% in poorer communities. The success of cereals is attributed to their high yields, adaptability to diverse environments, and ease of storage and processing. In the world market, approximately 42% of total cereals consumed are from wheat, rice, and maize (FAO 2018). However, there are a variety of other cereal competitors growing in popularity to combat climate instability. *Sorghum bicolor* (sorghum) is the fifth most important cereal crop worldwide and is an important cereal crop for nutritional food security. It is often grown on marginal dryland cropping areas due to its ability to thrive on marginal soils in hot and arid conditions,

producing more biomass per unit water than other cereals (Steduto et al. 1997; Kim and Day 2011). Heat and drought tolerance are becoming increasingly important with the issues with climate instability, unpredictability of rain fall, and the growing human populations.

Sorghum is mostly grown as a food and feed crop, although in comparison is regarded as having low calorific and nutritional value, both of which are influenced by grain protein quality. Most cereal grains often have a total protein content of approximately 10–14%, but sorghum has a more diverse range between 6 and 18% and averaging around 11% for elite breeding lines (Lasztity 1996). In cereals, the composition and interactions of the protein–starch matrix within the endosperm are often the determinants of its commercial value (Jampala et al. 2012). The majority of grain storage proteins accumulate in membrane-bound, spherically shaped, specialised organelles called the protein body (Holding 2014). These protein bodies are embedded within a matrix surrounding the starch granules, which are tightly packed in the vitreous endosperm and loosely packed in the floury endosperm (Pedrazzini et al. 2016). The proteins can be categorized based on physiochemical properties and separated into three main groups: albumins/globulins (water/salt-soluble), glutelins (acid/alkali-soluble), and prolamins (alcohol-soluble) (Osborne 1907, 1909; Taylor and Schussler 1986). The albumins and globulins are primarily in the pericarp and embryo of the grain and have been reported to comprise 10–30% of the total grain protein in the majority of cereals, although in rice and oat, they are the major endosperm storage protein (Shewry and Halford 2002; Afify et al. 2012). Glutelins are the major components of the complex lattice network that surround the starch granules where the protein bodies are embedded (Beckwith 1972; Subramanian et al. 1990). The prolamins make up over half of the total grain protein and prolamins polymers are almost exclusively housed within the protein bodies (Belton et al. 2006). In the endosperm, 77–82% of the proteins belong to the kafirin family which are abundant in glutamine and proline residues and highly hydrophobic with complex tertiary structures (Shull et al. 1992).

The kafirins are classified into various subclasses; α -kafirin (22–26 kDa), β -kafirin (18 kDa), γ -kafirin (28 and 50 kDa), and δ -kafirin (13 kDa). They are synthesized at the surface of the rough endoplasmic reticulum (ER) and packaged, so the most abundant α -kafirin is in the interior and β - and γ - are externally located, encapsulating the α -kafirin (Shull et al. 1992; Oria et al. 2000). The δ -kafirin has been found only at very low levels in the endosperm, reported to be in the core of the protein body and comprising less than 1% of total kafirins (Izquierdo and Godwin 2005; Mokrane et al. 2009; Cremer et al. 2014a). There are variable levels of kafirin accumulation throughout the endosperm. The vitreous endosperm is abundant in α -kafirin, comprising almost

80%, with the peripheral kafirins comprising the remainder. However, the floury endosperm has an increased level of the β - and γ -kafirins reaching 12 and 20%, respectively, meaning that the α -kafirin is reduced to only ~70% of the total kafirin content within the floury endosperm (Watterson et al. 1993).

The β - and γ -kafirins in the periphery of the protein bodies are linked to the reduced digestibility of the grain. They both contain high levels of cysteines, leading to increased inter- and intra-disulphide linkages which causes the outer shell of the protein body to be highly resistant to protease degradation (Duodu et al. 2003; Nunes et al. 2004; Belton et al. 2006). The low digestibility of the grain facilitates sorghum's low calorific value, which is further impacted by cooking (Axtell et al. 1981; Hamaker et al. 1986; Duodu et al. 2003). Sorghum flour is also low in the essential amino acids, such as lysine, threonine, and methionine, and can be high in condensed tannins that further reduce absorption of nutrients. Efforts to improve the end-quality of sorghum germplasm have led to the development of highly digestible sorghums with improved soft endosperm qualities, but these lines are often linked to increased susceptibility to environmental stresses, including moulds and droughts, and thus are underutilised (Rooney 1996). Therefore, there is high demand for the development of sorghum lines with improved grain protein.

A number of natural β -kafirin-null mutants have been cultivated in regions of northern Nigeria. These are favoured for a specific porridge renowned for its desirable consistency, known as *ogi*, due to its improved mouth feel. It has been shown that these mutants have higher protein digestibility (Laidlaw et al. 2010; Cremer et al. 2014a, 2014b). A selected variant was integrated into the Queensland breeding program, giving rise to the improved inbred QL12. Its enhanced quality was attributed to its β -kafirin allele with a cytosine insertion at position +206, causing a frame-shift mutation and early stop codon, which makes the protein unstable, as demonstrated by the lack of detectable levels in the mature grain (Cremer et al. 2014b). Phenotypically, these lines have increased grain protein levels with improved digestibility, although they have similar levels of cross-linking within the endosperm proteins. Unfortunately, lines with this β -kafirin allele also demonstrate reduced field emergence, which is not uncommon in cereals with improved protein digestibility (Han et al. 2012). Similarly, studies show that α -kafirin knockdown and knockout lines lead to improved digestibility but showed opaque endosperm with reduced germination efficiency (Kumar et al. 2012; Li et al. 2018). These negative pleiotropic effects can potentially be remedied through other mutations, as demonstrated in improved Quality Protein Maize (QPM) (Pixley and Bjarnason 2002; Maqbool et al. 2021).

This study used CRISPR/Cas9 gene editing technology to target the β -kafirin gene in the transformable line Tx430.

This technology has already led to breakthroughs in grain quality improvement through targeting the α -kafirin alleles (Li et al. 2018), and by knocking out the other prolamins, we can gain insight into the grain proteome in mature grain and the downstream impacts on quality. If β -kafirin knockouts can be shown to recapitulate the improved digestibility of null lines, this could be a simple means to improve the utilisation of sorghum for human and animal nutrition. However, this suggests that this trait is not as simple as previously believed, and the loss of this kafirin may lead to alternate traits in certain genetic backgrounds by the naturally occurring null lines. Instead, in certain genetic backgrounds, there was a compensatory mechanism with the peripheral kafirins where a significant increase in γ -kafirin levels has led to no changes in grain quality. This compensation was not found in the null mutant lines, which suggests that γ -kafirin may be able to compensate for the loss of β -kafirin in some genetic backgrounds through an unknown mechanism and is likely the cause of the changes to vitreous endosperm morphology that is likely linked to the variation in digestibility of the grain.

Materials and methods

Plant material

Transformation of *Sorghum bicolor* (sorghum) was performed via microprojectile bombardment of embryogenic callus generated from the Tx430 variety to create the gene edited lines following previously developed protocols (Liu et al. 2014). The sorghum variety Tx430 was provided by Queensland Department of Agriculture and Fisheries (QDAF). Outcomes were previously described by Massel et al. (2022), where five transgenics were produced targeting the β -kafirin gene (*Sobic.009G001600*), four of which were homozygously edited: *bkaf_1*, *bkaf_3*, *bkaf_4*, and *bkaf_5*. The types and consequences of the edit are described in the results. These gene edited (GE) lines were also compared to known β -kafirin-null lines, QL12 and Tx2737 (provided by QDAF) which are derived from a common ancestor sharing an SNP within the β -kafirin gene (described in Results) and have been shown to lack the protein in the mature grain (Laidlaw et al. 2010; Cremer et al. 2014a).

Sequence analysis gene editing

DNA was extracted from putative transgenics to verify the β -kafirin allele via PCR with MyTaqHS (Bioline) following the methodology outlined in Massel et al. (2022). All PCR products were cleaned using ExoSAP-IT (Thermo Fisher) following the manufacturer's instructions and sent

for Sanger sequencing using both forward and reverse sequencing for the β -kafirin allele. To assess the potential functional disruption of the gene editing, the programs Phyre2 (Kelley et al. 2015) and SWISS-Model (Guex and Peitsch 1997; Waterhouse et al. 2018) were performed protein modelling on the β -kafirin amino acid sequences.

Glasshouse trials

Once the plantlets had developed strong roots in tissue culture, the T0 plants were moved to a temperature-controlled glasshouse (28/20 °C) at The University of Queensland in Brisbane, Australia. They were planted two per 18L ANOVA[®] pot in potting mix and maintained with fertiliser treatment as necessary until maturity. Leaf segments were taken at 5-leaf stage for genotyping analysis of the β -kafirin gene and presence of Cas9 gene integration. Mature grain was harvested, dried, and analysed as described below in later sections.

A glasshouse trial was performed to phenotype the T1 generation of the β -kafirin GE lines to the wild-type Tx430 genotype. These experiments were performed in temperature-controlled (28/20 °C) facilities at The University of Queensland. Four biological replicates of three independently edited β -kafirin lines and a Tx430 control were grown in potting mix in 18L ANOVA[®] pots in a randomised complete block design. These were planted during the winter and sprayed weekly with 0.01 M CaCl₂ to reduce calcium deficiency and treated on an as-needed basis with other nutritional supplements until maturity. To verify heritability, leaf segments were taken at 5-leaf stage and genotyped for confirm the presence of the β -kafirin edit. Mature grain was harvested, dried, and analysed as described below in later sections.

To compare these traits to the null lines, a glasshouse trial was performed on the T2 generation of two GE lines (*bkaf_1* and *bkaf_4*), two null lines (QL12 and Tx2737), and wild type (Tx430) in a randomised complete block design. These six genotypes were grown in temperature-controlled (28/20 °C) glasshouses in the St Lucia campus facilities at The University of Queensland. In this experiment, the germplasm were sown in 4L ANOVA[®] pots in potting mix and treated with nutrients on an as-needed basis until maturity. For gene expression analysis, additional plants were sown, three plants of the same genotype per 18 L ANOVA[®] pots and treated as above. Mature grain was harvested, dried, and analysed as described below in later sections. Using the same grain, a separate germination test was performed in a growth chamber (28/20 °C) where 100 seeds were sowed in seedling trays with potting mix. After 2 weeks, seeds that germinated were counted.

Crude grain protein analysis

To determine protein content, 2 g of seeds were randomly selected from each harvested panicle and ground into flour using the Retsch Ball and Mill set with a TissueLyser (Qia-gen) for 30.0 Hz for 2 min. For each sample, 100 mg of flour was analysed for nitrogen by combustion using the LECO 928 analyser where the crude protein conversion factor of 5.6 was used to calculate protein content (%) (Mariotti et al. 2008).

Similarly, *in vitro* digestibility assays were performed on separate 100 mg aliquots of flour samples. Both pepsin and chymotrypsin were used to assess digestibility separately following the procedure outlined in Liu et al. (2019) with two technical replicates. For the pepsin digestion, 100 mg of flour sample was incubated for 8 h on a shaker at 37 °C in 15 mL of 0.1 M acidic phosphate buffer solution (pH 2.0) with 1.5 mg mL⁻¹ pepsin (Thermo Fisher) comprised of KH₂PO₄. After 8 h, the reaction was stopped by adding 1 mL of 2 M NaOH to the solution and vortexed. The flour samples were then washed in the phosphate buffer without pepsin enzyme three times, removing the supernatant each time, and dried overnight at 70 °C before analysis. For the chymotrypsin digestion, 100 mg of flour sample was subjected to 8 h shaking incubation at 35 °C in 15 mL of alkaline digestion buffer [100 mM Tris-HCl, 10 mM CaCl₂ (pH 7.8) with 1.5 mg mL⁻¹ chymotrypsin (Thermo Fisher)]. After 8 h, the reaction was stopped by heating the reaction to 60 °C for 20 min with intermittent vortexing. The samples were then washed three times with digestion buffer without the chymotrypsin and dried at 40 °C overnight. The digested flour samples were then analysed for nitrogen content as outlined previously.

Grain protein extraction and SDS-PAGE

Aliquots of the flour samples were used to sequentially extract seed storage proteins. Proteins were extracted on all GE lines, as well as an internal transgenic control (no editing), non-transgenic Tx430, and the natural β -kafirin-null mutant genotype QL12. First, the albumins and globulins were extracted using water/salt extraction solvent: 50 mM Tris-HCl (pH 7.8), 100 mM KCl and 5 mM EDTA. Flour samples (100 mg each) were suspended in 1 mL of this extraction solvent and vortexed for 5 min. The samples were spun down and supernatant containing the water/salt-soluble proteins were moved to a fresh tube. The process was repeated with 500 μ L of water/salt extraction buffer and supernatant pooled.

From the same flour pellet above, alcohol-soluble proteins were extracted containing the prolamins. The prolamins extraction buffer contained 60% tert-butanol, 0.5% sodium acetate (w/v) and 2% β -mercaptoethanol (v/v). Again, 1 mL

of solvent was added to the flour sample and vortexed for 5 min. The sample was spun down and supernatant containing the soluble proteins transferred to a fresh tube. The process was repeated with 500 μ L of prolamins extraction buffer where the supernatant was pooled.

Finally, polymeric proteins were extracted from the pellet, containing glutelins and residual proteins. These were extracted using an alkaline extraction buffer containing 12.4 mM sodium borate (pH 10), 2% SDS (w/v) and 1% β -mercaptoethanol (v/v). The flour samples were vortexed for 5 min in 1 mL of the residual protein buffer and then spun down and the supernatant transferred into a fresh tube. The extraction process was continued with 500 μ L of buffer and supernatants pooled.

A one-dimensional SDS-PAGE was performed for each extraction. For the albumin/globulin and glutelin/residual protein samples, 10 μ L of supernatant was resuspended in 5 μ L of 3X loading buffer [100 mM Tris-HCl, 1% SDS (w/v), 0.01% bromophenol blue, 20% glycerol, and 0.05% β -mercaptoethanol (v/v)]. The samples were heated to 95 °C and then chilled on ice prior to being loaded onto small format precast any KdTM (10–150 kDa range) gradient gels (Bio-Rad). An increased ratio of loading buffer was required for the prolamins extracts to achieve the appropriate density. For these samples, 5 μ L of a sample was mixed with 10 μ L of 3X loading dye before being heated to 95 °C and chilled.

For each gel, 10 μ L of BenchMark™ Pre-stained protein ladder (Life Technologies) was also run. Gels were run in SDS running buffer [25 mM Tris, 200 mM glycine, and 0.1% SDS (w/v)] at 200 V for approximately 35 min or until the loading dye front reached the bottom of the gel. The gels were washed three times in ddH₂O, stained in Bio-Safe Coomassie blue for 1 h on a shaker, and then de-stained by washing three-to-five times for 15 min each with ddH₂O before being photographed.

Quantitative proteomics

Proteomics was performed using liquid chromatography–mass spectrometry following the protocol outlined by (Kerr et al. 2022). Here, 50 mg of sorghum seeds were ground and solubilized in 600 μ L of denaturation buffer (6 M guanidine HCl, 50 mM Tris HCl buffer pH 8, and 10 mM DTT), and incubated for 30 min at 30 °C with shaking. To alkylate the cysteines, acrylamide was added to a final concentration of 30 mM and incubated for 1 h at 30 °C with shaking. Excess acrylamide was quenched with an additional 10 mM DTT, and samples were centrifuged at 18,000 g for 10 min. Proteins from 50 μ L of supernatants were precipitated by the addition of 1 mL 1:1 methanol:acetone and incubation at – 20 °C overnight. After centrifugation, precipitated proteins were resuspended in 100 μ L of 100 mM

ammonium acetate with 1 µg trypsin (proteomics grade, Sigma) and digested for 16 h at 37 °C with shaking.

Peptides were desalted with C18 ZipTips (Millipore) and measured by data-dependant acquisition (DDA) and data-independent acquisition (DIA) by LC–ESI-MS/MS using a Prominence nanoLC system (Shimadzu) and TripleTof 5600 mass spectrometer with Nanospray III interface (SCIEX) as previously described (Xu et al. 2015).

Peptides and proteins were identified as previously described in Kerr et al. (2022) using Protein Pilot 5.0.1 (SCIEX). Sorghum samples were searched against all predicted proteins from *Sorghum bicolor* v3.1.1 (PRJNA3869, downloaded 29 May 2017; 47,205 proteins). The search database also included predicted edited sequence from transgenic lines and contaminant proteins (custom database; 298 proteins). Search settings were: sample type, identification; cysteine alkylation, none; instrument, TripleTof 5600; species, none; ID focus, biological modifications; enzyme, trypsin; search effort, thorough ID.

The abundance of peptide fragments, peptides, and proteins was determined using PeakView 2.2 (SCIEX) with settings: shared peptides, allowed; peptide confidence threshold, 99%; false discovery rate, 1%; XIC extraction window, 6 min; XIC width, 75 ppm. Protein-centric analyses were performed as previously described (Kerr et al. 2021). The proteins were normalised in each sample to the sum of abundance of all proteins from that sample. Analysis of variance (ANOVA) was conducted on the normalised abundance following by a Tukey post hoc test to determine significance between genotypes (GraphPad Prism version 8.2.1).

Physiological data

Measurements included plant vegetative height, plant height, panicle length, and tiller number. Biomass and panicles were dried at 40 °C for 1 week and weighed. Grain traits, such as thousand kernel weight (TKW), total seed count, yield of main tiller, total yield, and the harvest index, were calculated. Grains were cut medial-longitudinally with a razor blade and photographed using a dissecting microscope. Analysis of variance (ANOVA) was conducted on all the harvest data following by a Tukey post hoc test to determine significance between genotypes (GraphPad Prism version 8.2.1).

Gene expression analysis

This was performed on developing grain from plants grown special for this assessment as to not effect yield analysis for other experiments. Three plants per 18 L ANOVA[®] pot were sown in potting mix for each of the

T2 lines tested (WT, *bka1_1*, *bka1_4*, QL12, and Tx2737). For each panicle on the main stem, flowering time was recorded at the ‘half bloom’ stage. Developing grains were sampled at 14 days after pollination (DAP) and snap frozen in liquid nitrogen and stored at – 80 °C. The tissue was ground into a fine powder using a mortar and pestle under liquid nitrogen before undergoing the RNA extraction protocol for starchy cereal grains (Wang et al. 2012). The extracted RNA was reverse transcribed using GoScript[™] Reverse Transcription System (Promega) following the manufacturer’s instructions and stored at – 20 °C.

For qPCR, the reference gene selected for this study was Ubiquitin (*Sobic.009G174900*), orthologue of the gene identified in rice for normalisation in qPCR, and gene-specific primers were synthesized for the reference gene and the two peripheral kafirins, β-kafirin (*Sobic.009G001600*) and γ-kafirin (*Sobic.002G211700*) (Jain et al. 2006; Table S1). The qPCR reactions used GoTaq[®] qPCR Master Mix (Promega) according to the manufacturer’s instructions (10 µL per well), and then pipetted into a Hard-Shell[®] 96-Well PCR plate and sealed with Microseal ‘B’ PCR plate sealing film (Bio-Rad). CFX96 Touch Real-Time PCR Detection System (Bio-Rad) and programmed with the following reaction conditions: 95 °C for 3 min followed by 45 cycles of 95 °C for 10 s, 64 °C for 10 s, and then 72 °C for 30 s with a fluorescence reading taken at the end of each cycle. A melt curve protocol took place directly after cycling with the following conditions: 95 °C for 10 s, 60 °C for 1 min, and then a ramp up of 0.5 °C s^{–1} to 95 °C. The qPCR data were then analysed to determine the mean PCR efficiency (PE) of each amplicon using the software LinReg (Ruijter et al. 2009). Relative expression was calculated by dividing the PE-Cq of the gene of interest by the reference gene PE-Cq. A Levene’s test was conducted on the relative expression data to determine the equality of variances, and then, an ANOVA was test conducted (GraphPad Prism v8.2.1).

Scanning electron microscopy

Scanning electron microscopy (SEM) was used to assess physical endosperm characteristics in mature sorghum grains from T2 yield experiment with Tx430, edited *bka1_1* and *bka1_4* lines, as well as the null QL12 and Tx2737 lines. Grains were cut medial-longitudinally with a razor blade and mounted on SEM pedestals with double-sided carbon tapes. After sputter-coating with platinum using the Safematic CCU-010 Compact Coating Unit, the samples were observed with a Hitachi TM4000 scanning electron microscope at 11.5 mm and 10 kV. Photographs were taken at 11.5 mm working distance with a 10 kV beam acceleration at either × 500 or × 1500 magnification.

Results

Gene editing of β -kafirin

CRISPR/Cas9 gene edited lines were observed to have an insertion of a nucleotide at position + 363nt within the β -kafirin gene sequence. Rather than generate an early stop codon, the frame-shift mutation results in a protein with an extended peptide sequence (Fig. 1a). The indel occurs on the 113th amino acid extending the protein from a length of 192 amino acid residues to a protein that contains 252 amino acids. The amino acid sequence is considerably different from the 113th amino acid onwards in comparison to Tx430. Whereas the allele in both the β -kafirin-null lines has a truncated protein of only 70 amino acids due

to the mutation occurring on the 69th amino acid (203rd nucleotide). Based on the protein modelling performed, the predicted functional region of this protein codes for a bi-functional (α -amylase/trypsin) inhibitor/seed storage 2S albumin protein, which is disrupted by both the CRISPR/Cas9 edit and in the null line mutation (Fig. 1b).

To quickly verify the lack of β -kafirin protein, SDS-PAGE was performed on the alcohol extract from mature grain samples. β -Kafirin is visualised at 18 kDa in the Tx430 and tissue culture control samples from the alcohol (prolamin) extract, whereas there is an absence of a band in the GE lines (Fig. S1). The α - and γ kafirin co-localise and are detected at 26 kDa in all the samples, whereas the δ -kafirin would appear around 13 kDa. A new band appears in all of the edited samples at approximately 10 kDa. In the other protein extracts, there were minimal visible differences

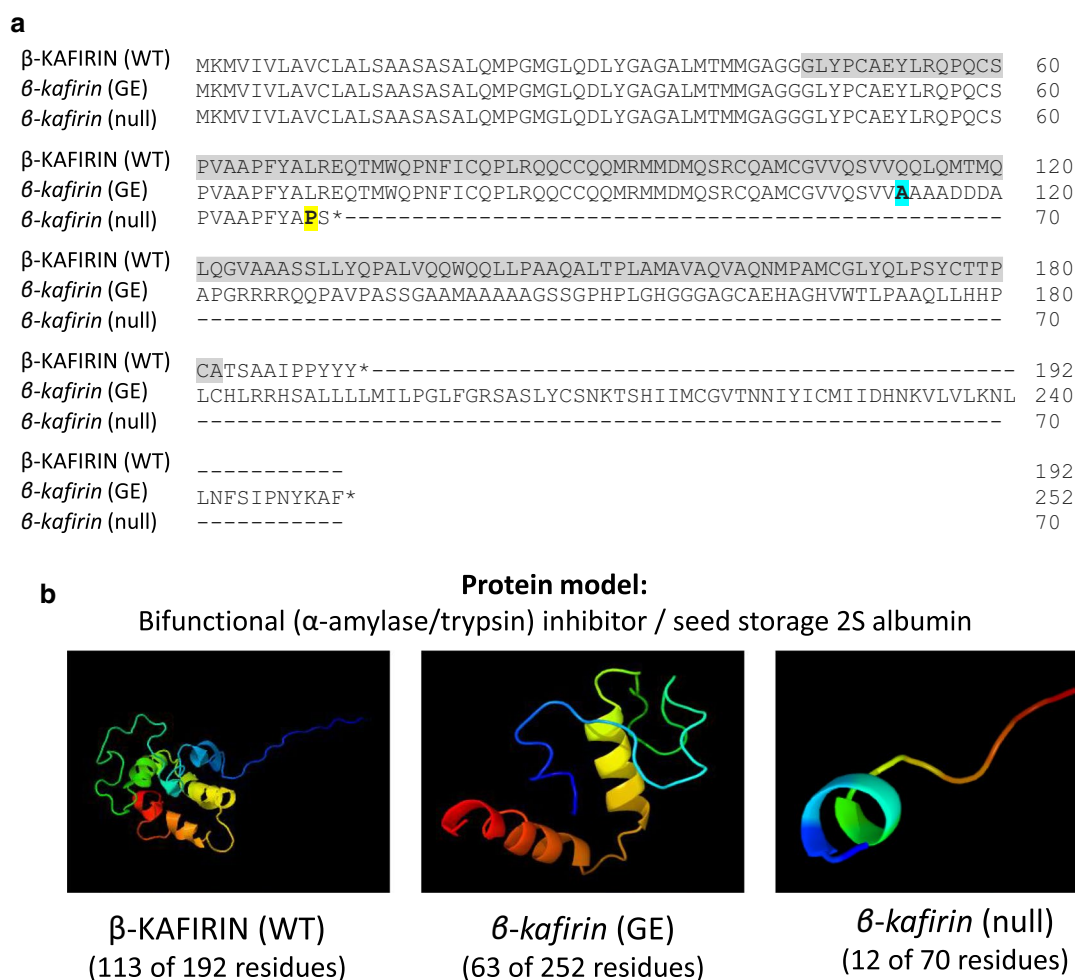


Fig. 1 Comparison of the WT protein, *bkaf* gene edited protein, and null β -kafirin proteins. **a** Clustal-Omega alignment amino acid alignment of the β -kafirin protein. Blue colour highlights the first amino acid affected by the insertion of a nucleotide through gene editing, whereas yellow colour shows the first amino acid affected in the null mutation. Highlighted in grey colour is a protein template for a bi-

functional (α -amylase/trypsin) inhibitor/seed storage 2S albumin protein. **b** Protein modelling through Pyre2 of the expected outcome to the motif highlighted in green. Shown are the β -kafirin model for WT (99.5% confidence), the Gene Edited (GE) variants (98.2% confidence), and null variant (14.9% confidence)

detected (data not shown). Proteomics and gene expression are described later to further validate the lack of β -kafirin in the GE lines.

Loss of β -kafirin in Tx430 does not impact protein content

The T0 lines were analysed for grain quality traits to give an indication of grain quality and size outcomes, even though there is only one representative line per edit. Yield parameters were not measured as these plants came directly from tissue culture and show phenotypes more likely related to synthetic auxins from the media. The quality results from T0 lines were found to corroborate those found when testing the next generation as described below (data not shown). When assessing the T1 generation, the lack of β -kafirin had no impact on yield ($P=0.9879$) or seed number ($P=0.9353$). TKW was trending to be slightly increased in the edited varieties with average weights ranging to be on average 2.3–6.0 g heavier, although somewhat more variable (Fig. 2a). Grain number and size were compared directly and found to show a similar negative correlation (Fig. S3a).

The GE lines were found to have similar levels of total grain protein with averages of 14.9, 15.9, and 15.2% for *bka1_1*, *bka1_3*, and *bka1_4*, respectively, in comparison to the 16.7% protein content of the WT grain (Fig. 2b). The GE lines were trending to have a slight reduction in total grain protein content, although this was not statistically significant ($P=0.113$, 0.704 , 0.20 for *bka1_1*, *bka1_3*, and *bka1_4*, respectively). In vitro digestibility assays showed a trend for pepsin digestibility to be reduced in the gene edited lines ($P=0.211$, 0.110 , 0.087 for *bka1_1*, *bka1_3* and *bka1_4*, respectively) (Fig. 2c). The results were more variable in the chymotrypsin assay and did not show a common trend across all of the GE lines, as *bka1_1* is trending to be less digestible ($P=0.091$), whereas *bka1_4* was trending to be more digestible ($P=0.64$).

β -Kafirin knockout has no impact on physiological development

From both T1 and T2 glasshouse trials on the GE lines, there appeared no evidence of physiological changes to development in the GE lines in comparison to the Tx430. Plant height, panicle length, or canopy biomass showed no significant changes (Fig. S2a, S2b). However, there are drastic differences in the overall physiology in both the null lines compared to those with a Tx430 genetic background (Fig. S2a, S2b). In comparison to Tx430, QL12 is on average 27 cm shorter in plant height ($P<0.0001$). Whereas both the null lines had a shorter panicle by approximately 7.5 cm ($P<0.0001$) which averaged around 24.5 cm rather than 32 cm for those with a Tx430 genetic background. This reduction in panicle size has likely led to the decreased yield round in both the null lines, which was 13.7 g less in Tx2737 ($P=0.0258$) and 30.8 g in QL12 ($P<0.0001$), but when assessed for harvest index, these values were insignificant (Fig. S2c).

Germination rates were assessed through sowing 100 seeds in potting mix for each line. The results demonstrate that there was no impact on germination rates in the GE lines having an average rate of ~95%, similar to the wild type with no changes to vigour ($P<0.0001$). A common trait of the null varieties is low field emergence, and when tested in this system, they showed an average germination rate of 74% for QL12 and 68% for Tx2737.

Improved quality traits in null lines are not explained solely by lack of beta-kafirin

No significant difference in grain size or number was found when comparing the GE lines to the WT (Fig. 3a, b). The null lines did show some differences, where the grain from Tx2737 was significantly smaller the edited lines and WT grains, and the QL12 grains were

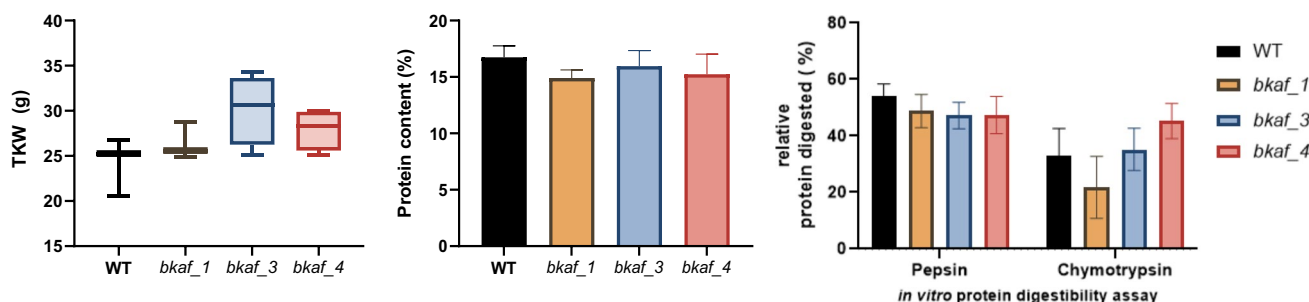


Fig. 2 Grain size and quality outcomes in the T1 GE lines compared to Tx430. Error bars represent SD ($n=4$). **a** Boxplot depicting the average Thousand Kernel Weight (TKW), where each sample was weighed three times and averaged ($n=4$). **b** Protein content (%) of mature flour ($n=4$). **c** Relative In vitro protein digestibility assays via

pepsin or chymotrypsin were used to determine the relative amount of protein digested in each sample. Four biological replicates were tested per line, with two technical replicates undergoing each digestion

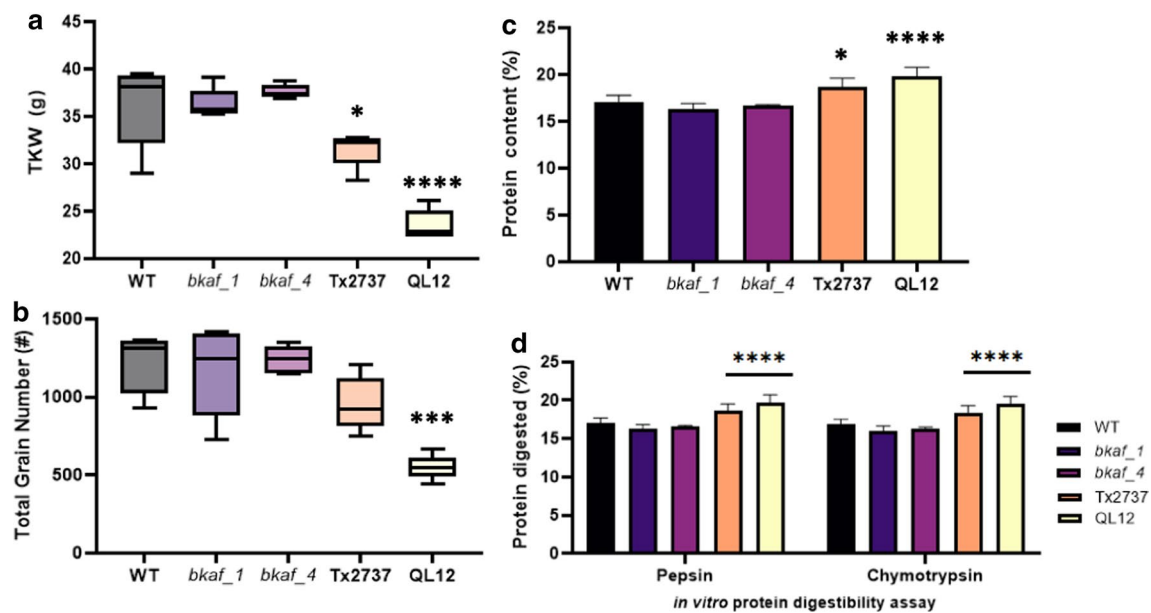


Fig. 3 Grain size and quality outcomes in the T2 generation comparing the three GE lines (*bkaf_1*, *bkaf_4*) with Tx430 and the two null cultivars (QL12 and Tx2737). For box plots and bar graphs, one-way ANOVA was performed in comparison to WT where error bars represent SD ($n=5$). Significance is shown as * is $0.01 < P < 0.05$, ** is $0.001 < P < 0.01$, *** is $0.0001 < P < 0.001$, and **** is $P < 0.0001$.

a Average Thousand Kernel Weight (TKW), where for each line where each sample was weighed three times and averaged. **b** Total grain number per plant as predicted from TKW values and total yield. **c** Protein content (%) of mature flour for each line. **d** In vitro protein digestibility assays via pepsin or chymotrypsin demonstrated the level of protein digested in each flour sample

significant smaller in size and weight than all of the lines including Tx2737 (Fig. 3a). Similarly, QL12 also showed fewer grains per panicle than the other lines tested. When comparing these values with their respective seed number, all lines showed TKW and grain number to be negatively correlated (Fig. S3b). Harvest index (HI) as calculated by yield (g) per total shoot biomass (g) remained similar across all of the lines (Fig. S3c).

Grain protein content values were found to be increased in the null lines, averaging 18.7 and 19.8% in Tx2737 and QL12, respectively, compared to 17.1% in WT (Fig. 3c). Although not significant, there is a trend for a reduction in protein content in the GE lines with averages of 16.3% and 16.6% in *bkaf_1* and *bkaf_3*, respectively. Quality was assessed through in vitro digestibility assays with pepsin and chymotrypsin and show the null lines to have improved digestibility (Fig. 3d). Since the lines have significantly variable protein content in the raw undigested flour, they were assessed based on the absolute amount of protein digested. In this assay, both the null lines show evidence of increased digestibility, whereas the GE lines have little-to-no impact. The similarity between the two null lines in their protein content and digestibility is likely due linkage drag that is associated with the identical ancestral mutation in the β -kafirin allele.

Lack of β -kafirin leads to a discrete grain proteome

Quantitative proteomics verified the knockout of β -kafirin protein in mature grain of GE lines. The GE lines form a distinct group from the Tx430 and QL12 lines. Principal component 1 describes 55.4% of the variation and differentiated between the lines that contain β -kafirin from those that lack it (Fig. 4a). Whereas the second component describes 22.1% and separates those from distinct genetic backgrounds, although the GE lines still remain separate to the Tx430 and tissue culture controls.

Heatmaps were generated to visualise this, which comprises every protein that makes up at least 1% of the total grain protein in each sample which include abundances for the kafirin proteins (Fig. 4b; Table S2). Alpha-kafirin (due to amino acid similarities, all are shown together in *Sobic.005G184800*) showed no significant difference between GE lines and wild type (Fig. 4c), while δ -kafirin (*Sobic.010g136100*) was not measured. Approximately 5% of the total proteome was made up of β -kafirin in WT and TC samples, whereas none was found in the GE lines or QL12 (Fig. 4c). The normalised abundance of γ -kafirin was similarly found to be around 6% of the proteome for WT and TC samples but was significantly increased and found to be doubled in the GE lines to approximately ~10% of the

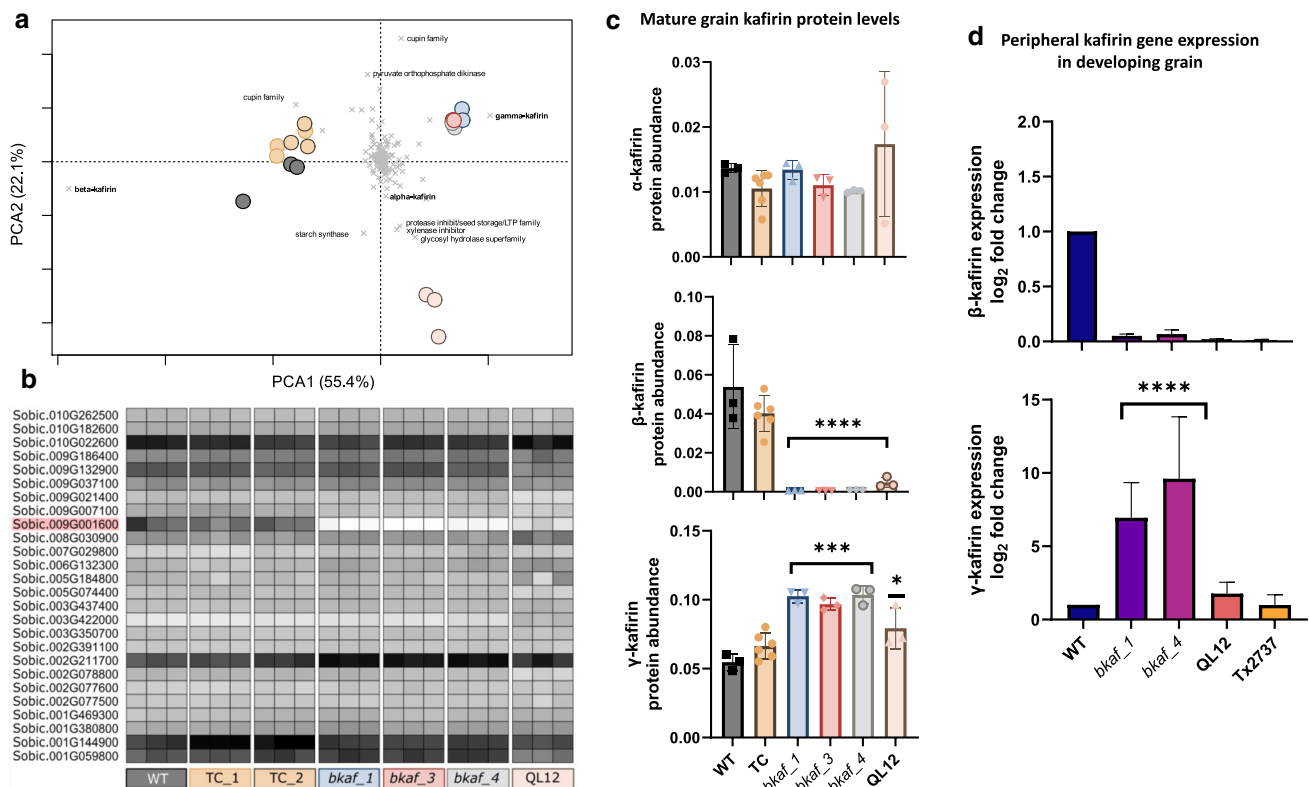


Fig. 4 Quantitative proteome and kafirin levels found across WT, GE, and null lines. For bar graphs, one-way ANOVA was performed in comparison to WT where error bars represent SD ($n=5$). Significance is shown as * is $0.01 < P < 0.05$, ** is $0.001 < P < 0.01$, *** is $0.0001 < P < 0.001$, and **** is $P < 0.0001$. **a** PCA of proteome of mature grain T0 samples comparing the normalised protein abundance. The addition of tissue culture controls (TC_1 and _2) has been added to ensure soma clonal variation did not affect pro-

teome. **b** Heatmap representing major proteins comprising at least 1% of the total proteome for each T0 sample. The β-kafirin protein (*Sobic.009G001600*) is highlighted in red. **c** Bar charts demonstrating changes in the normalised α-, β-, and γ-kafirin levels across the T0 samples. For these graphs, the two TC samples were combined. **d** Gene expression of both peripheral kafirins (β- and γ-kafirin) in developing grain sampled at 14 days post-flowering in the T2 glass-house experiment

relative proteome (Fig. 4c). Similarly, in the QL12 line, the γ-kafirin abundance was found to comprise 8% of the total proteome. To verify changes in the abundance of γ-kafirin, gene expression level was assessed during grain development in the T2 generation experiment for null, GE, and WT lines. RNA was extracted at 14 days post-flowering where qPCR was performed for both β- and γ-kafirin expression (Fig. 4d). The lack of β-kafirin expression was verified in the null and GE lines which fits with lack of protein found in the mature grain. At this same time point, the levels of γ-kafirin showed a significant upregulation of γ-kafirin in the GE lines compared to WT, and this compensation was not as strong in either of the null lines tested.

Other major grain proteins outside of the kafirin family that are known to impact quality were analysed to check for changes. Those involved in protein processing pathways were considered, although it is difficult as many of these genes are uncharacterised in sorghum. Chaperone proteins, such as Heat Shock 70 proteins (hsp70; *Sobic.004G011700*

(BiP) and *Sobic.003G350700*) and Heat Shock 90 (hsp90; *Sobic.010G267400* (GRP4)), are common indicators of ER stress and were in similar concentrations across the lines. Additionally, an endosperm-abundant protein disulphide isomerase (PDI; *Sobic.005G074400*), also involved in protein folding and can be an indicator of ER stress, was significantly increased in the GE lines (avg ~ 1.16% total protein) compared to Tx430 (~ 0.96% total protein) but was significantly reduced in QL12 (0.66%). Protein peptidyl-prolyl cis/trans isomerases (PPIase; *Sobic.004G018400*), which are known to have roles in grain quality, demonstrated similar trends with increased levels in GE lines and a reduction in QL12.

Proteins involved in starch biosynthesis pathway were also assessed (Campbell et al. 2016). Most of the proteins were undetected in the mature grain. Importantly, the only protein found in abundance was Granule-Bound Starch Synthase 1 (GBSS1 or Waxy; *Sobic.010G022600*) protein, which was found to be reduced by ~ 0.5–0.75-fold

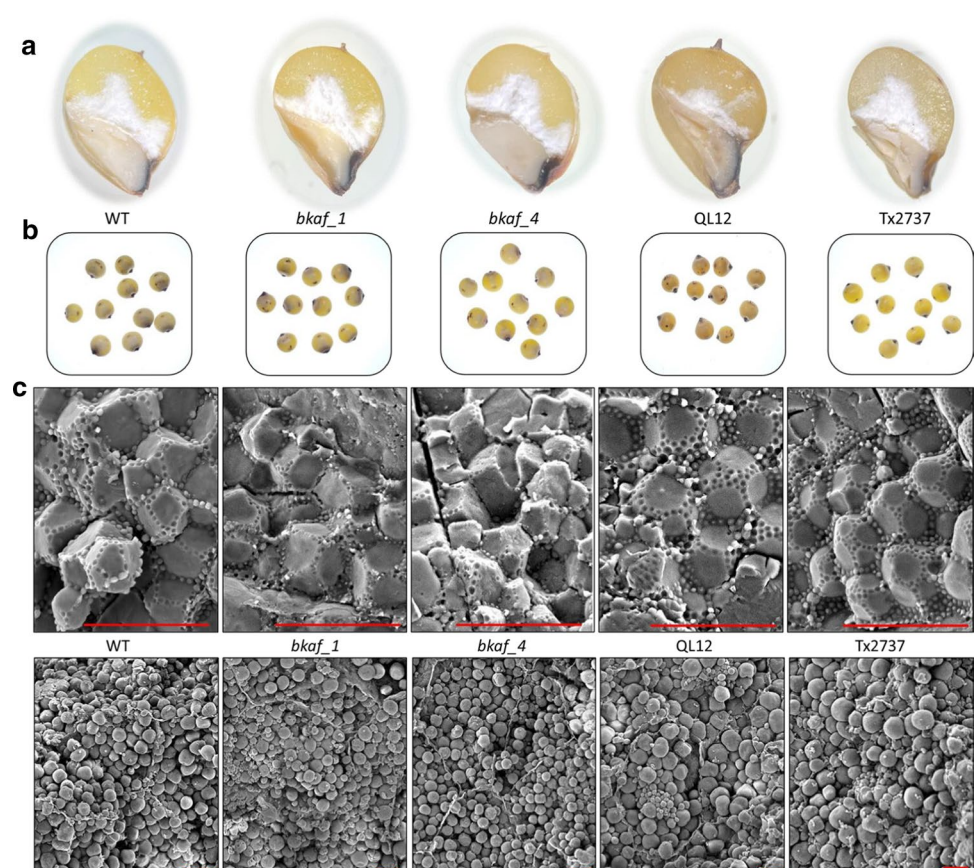
compared to the Tx430 lines. Of the other proteins detected, Starch Synthase IIA (SSIIa; *Sobic.010G09340*), Starch Branching Enzymes 2.2 (Amylose Extender 1 (Ae1); *Sobic.004G163700* & *Sobic.010G273800*), and Starch Phosphorylase 1 (PHO1; *Sobic.001G083900*) were found in very low quantities (less than 1%) and were not different across the lines tested.

Loss of β -kafirin linked to variation in mature grain morphology

The endosperm of the grain was visualised using light microscopy and SEM. Whole grains were visualised using a light box to determine opacity and were split medial-longitudinally to observe changes in floury and vitreous endosperm abundance and deposition (Fig. 5a). QL12 appeared to have a lower amount of floury endosperm found directly adjacent to the scutellum along the germ. Similarly, the floury endosperm of Tx2737 was found along the edge of the germ but in far more abundance. All of the GE lines and Tx430 showed similar distribution of an abundance of floury endosperm adjacent to the scutellum of the germ that has spread up toward the hilar region and reaching close to the pericarp. All of the above differences did not appear to have an effect on opacity of the whole grain.

Variation in the morphology of the endosperm becomes apparent in the SEM images (Fig. 5b). Overall, regardless of sorghum lines being visualised, the floury endosperm shows round starch granules with clusters of smaller protein bodies within a proteinaceous matrix. There does not appear to be much difference across all the lines. Whereas drastic differences can be visualised in the vitreous endosperm. In Tx430, polyhedral starch granules are surrounded by singular lines of rounded protein bodies. In both of the null lines, the ridges of protein bodies are often in multiple lines, showing increased number of protein bodies surrounding each starch granule. In both the Tx430 and null line vitreous endosperm, the granules appear polyhedral with rounded protein bodies; however, in the GE lines, the protein starch matrix looks aberrant. The protein bodies are irregularly shaped and are smaller on average. This shows that the starch granules are much more tightly packed within the protein–starch matrix in the GE lines and modify the appearance of the protein bodies.

Fig. 5 Representative samples of the grains visualised to assess mature grain phenotypes and changes in the endosperm for lines assessed in the T2 experiment. Includes WT, *bkaf_1*, *bkaf_4*, QL12, and Tx2737. **a** Grains per line were assessed using a dissecting microscope for changes in the relative levels of floury and vitreous endosperm. Grains were halved medial-longitudinally and imaged. **b** Ten grains of the same lines were imaged using a light box. **c** SEM images of the protein–starch matrix with photos of the vitreous endosperm (above) and floury endosperm (below) where the red bar is 30 μ m for all images



Discussion

β -Kafirin is found to be highly polymerised with intra- and intercellular bonds with γ -kafirin in the periphery of the protein bodies embedded within the protein–starch matrix, and has been demonstrated to be a significant factor contributing to sorghums low-protein digestibility based on the improved grain quality found in β -kafirin-null germplasm (Laidlaw et al. 2010; Cremer et al. 2014a). However, the link between the lack of β -kafirin and improved grain protein quality does not appear to be as straightforward as previously thought. We showed that knocking out β -kafirin through gene editing led to minor changes in protein content and had the tendency to reduce rather than improve digestibility. There were clear changes in the protein body morphology and structure within the protein–starch matrix in the vitreous endosperm. These outcomes are likely caused by significantly higher levels of γ -kafirin in the developing and mature grain. This compensatory increase in γ -kafirin levels is unfounded in the null lines through an associated linkage drag through an unknown mechanism.

In each of the GE lines, there was a one nucleotide insert into the β -kafirin gene causing a frame-shift mutation. Instead of an early start codon, it remains in-frame and translates into a protein of 252 residues. The mutation occurs within the 113th amino acid, and therefore, the overall protein is highly aberrant to the native one. This mutation is within the putative functional domain that spans residues 45–182. The lack of β -kafirin in the edited and null lines was verified using multiple approaches: limited expression was found in the developing grain, the band was absent in the prolamin extract on the protein gel, and the quantitative proteomics show virtually no levels of the protein in the mature grain.

No developmental changes were found in the GE lines, validating that β -kafirin has grain-specific roles, although previous research in overexpressing a modified β -kafirin gene had deleterious effects on seed set (Liu et al. 2019). The loss of this protein did not influence grain size or yield, nor did it impact germination rates. In terms of relative protein content, there was no impact by the loss of this protein which typically comprises 8–10% of the total kafirin content within the endosperm. These lines consistently had reduced protein digestibility, although it was more variable than the wild type and found to be not significant. These outcomes were surprising as β -kafirin-null varieties are known for their increased protein content and digestibility (Cremer et al. 2014a). The elevated levels of protein in the null lines are partially attributable to the high content and diversity of albumins and globulins (Cremer et al. 2014a). The increase of non-prolamin proteins

along with the lack of the highly indigestible β -kafirin was believed to be linked to their improved digestibility. However, it became evident that increasing grain quality in sorghum was more complicated than simply creating a non-functional β -kafirin protein.

The protein body contains two kafirins on its periphery which could mean that β - and γ -kafirin share some level of redundancy in their role as grain storage proteins. From the proteomics, the GE lines do appear to have a significant increase in γ -kafirin deposition in the endosperm, which was further corroborated through qPCR analysis during grain development. This study validated previous results that show null lines do not have compensatory levels of γ -kafirin (Elhassan et al. 2017). In maize targeting β - and γ -zein independently with RNAi showed no significant differences in grain phenotypes, but both peripheral kafirins were targeted there were drastic morphological changes in the protein bodies and opaque grain phenotypes (Wu and Messing 2010). In comparison to β -kafirin, γ -kafirin is more hydrophobic, has more cysteines, can form bonds with α -kafirin, and is believed to have roles in the initiation of protein body formation (Wu et al. 2013). Thus, increasing the levels of this kafirin may lead to higher levels of cross-linking, increased hydrophobicity, and increased number of protein bodies, all of which may contribute to the reduction in protein digestibility and morphological changes found in the vitreous endosperm. Improvement in grain quality in the null lines may be attributable to the overall reduction on peripheral kafirins which was not replicated by editing the β -kafirin allele in Tx430. To distinguish if the grain quality improvements in the null lines is independent of, or in combination with the lack of β -kafirin, one could express a functional β -kafirin allele using a transgenic approach to reduce the ambiguity this traits association due to linkage drag.

Among the sorghum prolamins, α -kafirin is the most digestible and is found in the highest quantities in the grain. Although there are 23 genes that code for this prolamin, they all are characterised under the code *Sobic.005G184800* and were found in similar levels in the GE lines of approximately 1.1% of the total proteome. This methodology is only semi-quantitative as the levels of kafirins are difficult to quantify using this LC–MS methodology. Similar to the reasons why these kafirins are difficult to digest as a food or feed, this approach relies on trypsin to breakdown the proteins into peptide fragments for analysis. Since the kafirins are highly polymerised (i.e. low solubility) and have very low numbers of protease sites, they are often digested into nonoptimal peptide fragments and under-evaluated in their relative abundance. Since the most indigestible kafirins are on the protein body periphery, accessing the internal core of the protein bodies is difficult and therefore impacts the levels of kafirins α - and δ -kafirin (Taylor and Schussler 1986; Watterson et al. 1993). Fluctuations in abundance are likely to be reflective

based on the strength of reproducibility across biological and technical replicates. Across the sampling, we found highly consistent results and were able to show dramatic differences in the GE lines to Tx430, as well as these lines to the null varieties highlighting the vast variation in the proteome found across sorghum genotypes.

Grain protein digestibility is a combination of many exogenous and endogenous factors that affect the levels of polymerization, cross-linking and development of the protein–starch matrix, and there are likely numerous attributes that influence the null lines' protein quality. When assessing the SDS-PAGE gels, apart from the obvious loss of β -kafirin in the prolamin extract, these GE lines also displayed a detectable band at approximately 10 kDa that was not visualised in the other samples. This may demonstrate an increase in abundance of either the δ -kafirin or a PPIase which are known to co-localise in the alcohol extract (Cremer et al. 2014a). Although δ -kafirin may not have been detected due to its limited abundance, we would predict to find even less in the GE lines. Previous studies have shown that β - and δ -zein are co-expressed, where it was suggested that β -zein is important for δ -zein stability in the protein bodies as it protects it from degradation (Bagga et al. 1997; Hinchliffe and Kemp 2002). Therefore, if sorghum's protein bodies function in a similar, one would predict a reduction in δ -kafirin in the GE lines through diminished expression and stability. Similarly, from the proteomics, there was only slight but insignificant increase in the levels of PPIase, so the small alcohol-soluble protein that appears in the GE lines remains uncharacterised.

Alternatively, there are numerous other factors that impact grain quality. Enzymes, such as thioredoxins, glutaredoxins, protein isomerases, PPIases, heat shock proteins, luminal BiPs, and α -amylase inhibitors, were found in similar abundances when comparing the GE lines to Tx430. The only significant factor was an endosperm-specific protein disulphide isomerase involved in modifying the cysteine residues where it was found in higher levels in the GE lines, suggesting that the GE lines may have increased levels of cross-linking. This protein has been shown to play roles in grain quality and indicates ER stress as it is upregulated by unfolded/mis-folded proteins (Wakasa et al. 2011; Han et al. 2012). Interestingly, the level of protein disulphide isomerase (PDI) is reduced in the QL12 line with improved grain quality in comparison to Tx430. Starch biosynthesis proteins were also considered, where only one Granule-Bound Starch Synthase (GBSS) was found to be reduced in the GE lines which may suggest that these lines would have reduced levels of amylose in their starch profile. Since starch can have dramatic effects on protein digestibility (Wong et al. 2009), it might be pertinent to test starch content and digestibility to understand the full impact of knocking out β -kafirin. This analysis was performed on mature grain so,

it is unknown if the relative levels of protein were different during endosperm development, so further assessment on developing grain would be necessary to validate any changes to their activity.

Visualising the grain and endosperm within these lines shows more evidence in changes in the development of the grain. Although the null lines have similar outcomes to opaque grain mutants with improved digestibility and poor field emergence, they do not have the true opaque endosperm phenotype (Mertz et al. 1964; Prasanna et al. 2001). In fact, there appears to be a reduction in the floury endosperm in both the null lines compared to the other Tx430 lines. Little impact is found in the floury endosperm, but changes are evident in the vitreous endosperm. The null varieties appear to have more and larger protein bodies surrounding their polygonal starch granules, which may provide increased accessibility within the vitreous endosperm for proteases. Conversely, the protein bodies in the GE lines appear irregularly shaped, small, and even damaged. The elevated levels of γ -kafirin could be a signal for the production of more protein bodies (Anyango et al. 2013; Wu et al. 2013). Without β -kafirin involvement in the cross-linking, the increased levels of intra- and inter-molecular disulphide linkages between the other kafirins are predicted as γ -kafirin is also capable of binding to α -kafirin. Further, heightened PDI levels are a signal of increased ER stress from mis-folded proteins, which may explain the morphological changes in the protein bodies in these GE lines (Gayral et al. 2017). Alternatively, these protein bodies could appear misshapen if the vitreous endosperm is too tightly packed between the starch granules, which could leave less space for proteases to break down the samples, revealing the trend for reduced protein digestibility. Regardless, it is evident that the loss of β -kafirin in the GE lines is modifying the protein–starch matrix in the vitreous endosperm and impacting the quality of the grain. Although the protein bodies must be the organelle effected by the loss of β -kafirin, these changes are having greater impacts on the organisational structure of protein–starch matrix endosperm which will, in turn, impact grain quality.

Conclusion

We have presented ample evidence to demonstrate that the improved grain protein quality within the null varieties cannot be directly attributed to the lack of β -kafirin. We showed that γ -kafirin will compensate for the loss of β -kafirin in some genetic backgrounds a role not observed in the β -kafirin-null lines such as QL12. Without an overall loss in peripheral kafirins, there are limited improvements to the grain quality, although visible changes in the vitreous endosperm were found, likely attributable to the increased levels γ -kafirin which is capable of more S–S linkages and

also can bind to α -kafirin. Further research could help disseminate the genetic link of improved grain quality in the null varieties, as it would be interesting to understand if those beneficial qualities are solely based on the reduced level of peripheral kafirins or if they have additional factors that are also linked to their improved quality. It may be that targeting γ -kafirin could have more significant effects on grain quality attributes.

Author contribution statement KM and IDG conceived and designed research. KM conducted gene editing and bulk of phenotyping. JH assisted in experiments comparing natural mutants to gene edited lines including qPCR. JR performed microscopy. EDK and BLS conducted, analysed, and assisted in the interpretation of the proteomics. KM wrote manuscript. All authors read, edited, and approved the manuscript.

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Data availability The datasets generated during this study are included in this published article and its supplementary information files.

Declarations

Conflict of interest The authors have no relevant financial or non-financial conflicts of interest to disclose.

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